Next.js Study Guide

Routing, rendering, and data fetching

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What is Next.js?

- Next.js is a React framework for building web applications. It adds routing, rendering options, data-fetching patterns, and production tooling.
- In the App Router, the app/ directory defines pages and layouts.

Routing basics

- app/page.js defines the home page at /.
- app/about/page.js defines the /about route.
- A folder such as app/resources/[id]/page.js can define a dynamic route.
- Use Link from next/link to navigate between internal pages.

Server and Client Components

- Components are Server Components by default in the App Router.
- Use 'use client' when a component needs browser APIs, state, event handlers, or effects.
- Keep components on the server where possible; use client components where interactivity is needed.

Data fetching

- Server Components can fetch data with async functions.
- Client-side libraries such as SWR can refresh browser data.
- Handle loading, empty, and error states clearly.

Good habits and practice

- Keep reusable UI in a components/ directory and shared static data in a separate module.
- Run npm run build before deployment and fix build errors.
- Practice: build a /resources page that maps over an array and displays resource cards.